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Modeling the Transport and Dispersion of Nitrate in Coastal Waters Using Advection-Diffusion Dynamics: The Deep Thinkers

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ABOUT

Our project will explore how ocean currents and turbulent mixing control the transport and dispersion of well-dissolved pollutants, specifically nitrate, in coastal regions. We plan to model this using a staged approach, starting with a simple 1D advection/diffusion setup and progressively adding complexity with a 2D flow and potential decay processes.

1. THE PROBLEM

1.1. *What is the question you are going to answer?*

The goal of this project is to investigate how nitrate spreads in coastal waters after entering from rivers. Specifically, we examine the horizontal transport of nitrate under the influence of ocean currents and diffusion processes. By simulating how nitrate concentration changes over time and space under specified boundary conditions, we can better understand the dispersion patterns of nitrate in coastal waters.

1.2. *Why is it interesting?*

Over the past few decades, the global demand for food has grown steadily, driving increased use of synthetic nitrogen fertilizers in agriculture (Singh and Craswell, 2021). While these fertilizers support higher crop yields, their excessive application has become a major source of nitrate pollution, which can leach into groundwater and flow into coastal waters, ultimately causing eutrophication (NRC, 2000; Howarth et al., 2005; Howarth and Marino, 2006). The resulting coastal eutrophication has intensified worldwide, leading to harmful effects such as algal blooms, oxygen depletion, and biodiversity loss in many marine ecosystems (NRC, 2000; Boesch, 2002; Heisler et al., 2008; Horta et al., 2021). The trends speak for themselves in that there is now an urgent need for monitoring and understanding nitrate pollution in coastal regions in order to assess environmental risks and inform mitigation strategies.

1.3. *How does it fit into known science?*

Nitrate transport in coastal waters has been continuously addressed by many previous studies due to its significant environmental impact (Eppley et al., 1979; Valiela et al., 1990; Sugimoto et al., 2009). To detect nitrate pollution and understand its behavior, scientists commonly conduct field observations, but these methodologies have unavoidable limitations such as high costs and operational constraints (Alahi & Mukhopadhyay, 2018). Numerical modeling is one of the methods that can complement these limitations, thereby widely applied to many research simulating how nitrate pollutants disperse under specific physical conditions (Eppley et al., 1979; Eslamian et al., 2023). The advection-diffusion equation is the mathematical base of this modeling, and its physical validity has been assessed and verified through comparisons with observational data (Walter, 2008; Ochwach et al., 2018). In this context, we are planning to move from a simple 1D model to a 2D simulation to better understand nitrate behavior in coastal waters, specifically how nitrate entering from rivers spreads under the influence of ocean currents and diffusion.

2. PROBLEM ORIGINATION

2.1. *Where did the problem come from?*

The core problem comes from the intersection of environmental monitoring and computational fluid dynamics (CFD). In reality, tracking the dispersion of well-dissolved pollutants like nitrate requires solving a highly non-linear coupled system. The fluid motion (flow field) is governed by the 3D incompressible Navier-Stokes equations (Kundu et al., 2018):

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f} \tag{1}$$

This velocity field u then drives the 3D advection-diffusion mass transport through the advection-diffusion equation, as given in Part 3. Because these coupled partial differential equations (PDEs) rarely have exact analytical solutions for complex, irregular coastal geometries, identifying the precise dispersion pattern relies heavily on numerical approximation and staged modeling (Zalesn, et al., 2017).

2.2. *Where did the idea come from?*

The specific idea to tackle this using a staged numerical approach originated from our recent course modules on numerical PDE solvers. Instead of diving directly

into the intractable 3D coupled system, we intend to validate our numerical scheme against classic mathematical baselines. For the initial 1D simplified case with initially clean domain with source $C(x,0) = \delta(x)$, we will compare our numerical output against the classic analytical solution for an instantaneous point source(Crank, 1979):

$$C(x,t) = \frac{1}{\sqrt{4\pi Dt}} \exp\left(-\frac{(x-ut)^2}{4Dt}\right) \tag{2}$$

By computing the error between our numerical results and this exact solution, we can verify mass conservation. Furthermore, the verified numerical laws and schemes will be indicators as we stage the problem and add complexity. We will also strictly monitor the Courant-Friedrichs-Lewy (CFL) condition to prevent non-physical oscillations(LeVeque et al., 1992):

$$C = \frac{u\Delta t}{\Delta x} \leq 1 \tag{3}$$

2.3. *Who have you discussed it with?*

To ensure the complexity of our project was appropriate, we discussed our initial concept with our instructor, Professor Allen. Initially, we considered importing complex, pre-generated flow fields from SalishSeaCast similar to a research done in 2022 (Mueller, 2022; Donaldson, 2025]. However, Professor Allen advised us to start with simple flow fields specifically generated for this project to better isolate and control the physical processes. This staged approach gives us the flexibility to manage the project's overall difficulty, while allowing us to systematically explore different aspects of numerical modeling. For instance, in our 2D stage, we plan to implement a classic boundary layer flow, where the current velocity u(y) follows a logarithmic profile relative to the distance from the shore (y)(Schlichting et al., 2016):

$$u(y) = u^* \ln\left(\frac{y}{y_0}\right) \tag{4}$$

By coding the flow field ourselves, we retain the flexibility to easily adjust, add, or remove flow complexity based on how our solver handles the numerical stability.

3. WHAT IS THE NUMERICAL PROBLEM?

We are going to solve the advection-diffusion equation for a passive tracer, as shown by (1):

$$\frac{\partial c}{\partial t} + \nabla \cdot (c\mathbf{u}) = D\nabla^2 c \tag{5}$$

Where c is our tracer concentration, u is the velocity field, and D is the diffusivity of our tracer. Our boundary and initial conditions specify the initial tracer concentration, and the concentration values at the sides of our domain. We will impose a no-flow boundary condition as shown below:

$$\frac{\partial c}{\partial x}|_{x=0,W} = 0, \frac{\partial c}{\partial y}|_{y=0,L} = 0, \frac{\partial c}{\partial z}|_{z=0,H} = 0 \tag{6}$$

Where W, L, H are the domain limits in the x, y, z directions respectively.

We will specify the initial condition as being a concentration of zero everywhere, and the boundary condition at the head of our domain will simulate the effects of a river inputting nitrate. At this point, we will specify the boundary condition as:

$$c(L,t) = c_R \tag{7}$$

Where c_R is a realistic river nitrate concentration.

And the initial condition is written as:

$$c(x,0) = 0 \tag{8}$$

We will specify our parameters of initial and boundary nitrate concentrations, as well as the nitrate diffusivity based on relevant literature. The velocity field will be specified based on typical current velocities observed in the region.

4. CAN YOU STAGE THE PROBLEM?

The problem can be staged to reduce the risk of model failure while building physical understanding incrementally. In the first stage we focus on the baseline one-dimensional horizontal advection-diffusion equation for nitrate concentration, given in §3 as Equation 5. At this stage, the initial condition is specified by Equation 8, $c(x,0) = 0$, which represents a system initially free of nitrate. The boundary conditions are defined by Equation 7 as the river input, $c(L,t) = cR$, which simulates the continuous introduction of nitrate from a river, and Equation 6 at the downstream boundary, $\frac{\partial c}{\partial x}|_{x=0,\,W=0}$ representing a no-flux condition. Flow is assumed to be steady and uniform (u = constant) and diffusivity is constant ($D = constant$). This simplified setup will allow us to verify the numerical stability, for instance, by checking the Courant-Friedrichs-Lewy condition, ensure mass conservation, and confirm that the plume is transported and spread correctly. Because the model is 1D and conceptually simple, it also allows for straightforward debugging.

In the second stage, we introduce additional physical realism by allowing the velocity field to vary spatially, $u(x)$, which can represent coastal currents or shear,

and potentially incorporating depth-dependent diffusivity, $D(x)$, to account for stratified turbulence. The initial and boundary conditions from Stage 1 remain the same, ensuring continuity. This stage will help us explore non-uniform flow and variable mixing after the plume shape while continuing to test the stability and accuracy of the numerical schemes under gradients in velocity and diffusivity.

5. OPERATIONS

We will organize the project with a combination of clear team roles, shared version control, and modular coding practices that will ensure that our collaboration is smooth:

5.1. *Code Sharing and Version Control*

All code will be stored in a shared **Github repository**, with pushes and pulls tracked to maintain version history. We will use **branches** for separate features or modules to minimize merge conflicts. For example, one branch may handle the 1D advection-diffusion solver while another handles visualization functions. Merge requests will be reviewed by at least one team member before integrating into the main branch. In push-comments, **state the reason for pushing and what you have done since the last push**.

5.2. *Division of Work*

- **Haechan (Coding Lead):** responsible for overseeing code development, ensuring functionality and coordinating Github updates
- **Aidan (Project Lead):** managing weekly goals, meeting agendas, ensures the team stays on schedule
- **Junqi (Research and Troubleshooting Lead):** formulates questions for instructors, proposes alternative modeling approaches if issues arise
- **Kat (Documentation Lead):** recording team discussions, tracks decisions, and ensures all tasks are logged for reproducibility. Responsible for ensuring all code is organized and properly documented before sending off to Haechan for reassurance in Github updates

The code will be divided into reusable functions and modules (i.e. advection solver, diffusion solver, boundary conditions, visualization). We will each develop and test modules independently then send them off for approval and integration into the final product. This will allow for parallel development without everyone working on the same file at the same time. Shared documentation in the repository will specify function inputs, outputs, and usage to make sure that our integration is smooth. While project roles are defined, all our members will contribute to literature review, data analysis, and report writing. Specific tasks (i.e. figures, tables, explanations) may rotate to ensure shared ownership while respecting individual strengths and uplifting the other team members.

5.3. *Handling Merge Conflicts*

We will minimize conflicts by ensuring each team member works on separate branches or modules whenever possible. If a merge conflict appears, the team member attempting the merge will communicate with others to resolve it collaboratively. Conflict resolution led by Git's merge tools (git merge –abort) and (git merge) and by discussion which version is correct. Resolved merges will then be tested immediately to ensure functionality is not broken. **Frequent commits and small merges** will help us reduce the likelihood of complicated conflicts.

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